**Code No: 113BP** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech II Year I Semester Examinations, April/May - 2023**
**DATA STRUCTURES**
**(Common to CSE, IT)**

**Time: 3 Hours** **Max. Marks: 75**

**Note:** i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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## PART - A

**(25 Marks)**

1.a) Describe the role of space complexity and time complexity in measuring the performance of a program. [2]
b) Differentiate between recursive and iterative algorithms. [3]
c) List the applications of stack. [2]
d) Differentiate Stack and Queue. [3]
e) Define a full binary tree and complete binary tree. [2]
f) List the applications of Trees. [3]
g) Define Hashing. [2]
h) State different types of collision resolving techniques. [3]
i) Compare binary tree and binary search tree. [2]
j) Write the properties of B-Trees. [3]

## PART - B

**(50 Marks)**

2. Discuss various the asymptotic notations used for best case, average case and worst case analysis of algorithms. [10]

**OR**

3. Write a program to insert an element in between two nodes in a double linked list. [10]

4.a) Explain the procedure to evaluate postfix expression.
b) Convert the following expression A + (B * C) - ((D * E + F) / G) into postfix form. [5+5]

**OR**

5. Implement a Circular queue of integer of user specified size and write the functions for initialize(), enque() and deque(). [10]

6.a) Discuss representation of binary tree.
b) Explain DFS graphs traversal algorithms with suitable example. [5+5]

**OR**

7.a) Explain with an example how to delete an element from maxheap.
b) Construct max heap for 150, 80, 40, 30, 10, 70, 110, 100, 20, 90, 60, 50, 120, 140, 130. [5+5]

8.a) Explain Binary search with example.
b) Explain the different hashing functions with an example. [5+5]

**OR**

9.a) State and explain radix sort with an example.
b) Apply heap sort on list of elements 14, 12, 9, 8,7,10,18,20,30. [5+5]

10.a) Explain the insertion operation on AVL trees.
b) Insert the following sequence of elements into an AVL tree, starting with an empty space: 10, 20, 15, 25, 30, 16, 18, 19 and delete 30 in the AVL tree that you got. [5+5]

**OR**

11.a) Explain Splay trees with example.
b) Construct a B-tree of order 3 with the following elements 25,10,20,30,80,40,50,60,82,70,90,85,93. [5+5]

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